



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGSD/ AGSRD Division

FIRST SEMESTER 2026-2027
COURSE HANDOUT

Date: 01.08.2026

In addition to part I (the General Handout for all courses, appended to the Timetable), this section provides further specific details about the course.

Course No. : **EEE/ INSTR F313**
Course Title : **ANALOG AND DIGITAL VLSI DESIGN**
Instructor-in-Charge : **ANU GUPTA**
Co-Instructor : **NEERAJ MISHRA**
Tutorial Instructors : **Neeraj Mishra ,Nitin Chaturvedi ,, Sambangi Ramesh , Rahul Kumar, Prashant Kumar, Govind Prasad**

1. Course Description:

This course introduces students to the principles and practical aspects of designing integrated circuits using CMOS technology. It covers both analog and digital VLSI design methodologies, exploring circuit-level building blocks, layout considerations, and performance trade-offs. Emphasis is placed on low-power, area-efficient, and high-performance circuit techniques, along with exposure to modern CAD tools and fabrication-aware design strategies. Students will gain experience designing critical components, including logic gates, adders, amplifiers, and mixed-signal interfaces.

2. Scope and Objective:

- The course is designed to familiarize students with different design principles used in the **Design of Digital & Analog VLSI Circuits & Systems** using CMOS logic.
- Understanding of **CMOS technology fundamentals** and their impact on design.
- Design and analysis of **digital logic circuits**: static, dynamic, sequential.
- Design and analysis of **analog CMOS circuits**: Op-Amplifiers, Advance current mirrors, Reference Circuit.
- Exploration of Technology **scaling trends, noise, and power-performance trade-offs**.
- Exposure to **EDA tools** for schematic capture, simulation, and layout.
- Bridging theory with practical insights via **Lab/Design assignments**.
- VLSI design activity will be understood by Synthesis and then Analysis at several successive levels of design abstraction.



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3. Text Book:

- (T1) Rabaey Jan M., Chandrakasan Anantha and Nikolic Borivoje, " Digital Integrated Circuits", Pearson Education. McGraw-Hill.
(T2) Behzad Razavi, "Design of Analog CMOS integrated circuits", Tata McGraw-Hill.

Prime reference Books:

- (R1) Kang. S.M and Leblebici Y., "CMOS Digital Integrated Circuits: Analysis and Design.
(R2) Johns. David A. and Martin K, "Analog Integrated Circuit Design," John Wily & Sons. Inc.

4. Reference Books:

1. Neil H.E. Weste and Kamran Eshraghian, "Principles of CMOS VLSI Design", Addison-Wesley Publishing Company.
2. Pucknell D.A. and Eshraghian K., "Basic VLSI design", Prentice Hall of India Pvt. Ltd.
3. Allen Phillip and Holberg Douglas, "CMOS Analog Circuit Design", Oxford University Press.
4. Gregorian R., Temes G.C., "Analog Mos integrated circuits for signal processing", Wiley Interscience Publication.
5. Sze S.M., "VLSI Technology", Second edition, TATA McGraw Hill.
6. Randall A Geiger, Phillips E. Allen, Noel R Strader, "VLSI Design techniques for analog and digital circuits," Tata McGraw-Hill.
7. Bhaskhar Jayram, "A VHDL PRIMER", Prentice Hall.
8. Palnitkar Samir, "Verilog HDL" Pearson Education Asia.
9. IEEE Journals of Solid State Circuits, IEEE Transactions on Very Large Scale Integration Systems.
10. Martin. Ken, "Digital Integrated Circuit Design", Oxford University Press.

5. Course Plan:

Module	Topics	Lecture Sessions	Reference	Learning Outcome
Module 1: Introduction to IC design	Introduction to VLSI Design Methodologies	2	Chapter-1 (T1) / (R1)	Knowledge of Existing Design Methodologies-Full custom/ semicustom, FPGA/ GA/ standard cell, Moors law
Module 2: Brief intro. to IC Fabrication	MOS Fabrication Process & technology, Circuit Layout & Design Rules, Layout, scaling techniques	3	Chapter-2,3 (T1), (T2) Reference book-7,8	IC Fabrication steps, CMOS process, Common centroid layout, multi-fingered gate layout technique
Module 3: Building Blocks of Digital VLSI	MOS inverter- D.C and switching characteristics	5	Chapter- 3, 4, (T1), R1	Knowledge of the Design of MOS Inverter



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Module 4: Design of Digital Logic Circuits in CMOS	1) Combinational static logic circuits	5	Chapter-5,6 (T1), R1	Design of Complex Digital Circuits with Time, Area, and Power Constraints
	2) Dynamic logic gates design styles	4	Chapter- 7,8,9 (T1), R1	Clocked circuit design styles, power/ delay constraints
	3) Timing, flip-flop, design	6	Chapter-10, 11 Chapter-10 (T1)+ (R1)	Clock time period, Impact of clock skew, clock jitter, static/ dynamic memory elements, buffered clock tree design
Review Session	Basic Analog CMOS Circuits	2	Chapter- 3,4,5,6 (T2)	Revision of Fundamentals of Analog CMOS Circuit design (single stage amplifier, differential amplifier, current mirrors)
Module 5: Design of Analog Building Blocks in CMOS	1) Advanced current sources & sinks; Reference circuit design) Advanced current sources & sinks; Reference circuit design Operational amplifier architectures, frequency compensation Techniques	6	Chapter-5,6 (T2), (R2)	Knowledge of the design of advanced DC Current/ voltage biasing circuits
	2) Operational amplifier architectures, frequency compensation Techniques	6	Chapter-9 (T2), (R2)	Knowledge of the design of stable single & multi stage amplifier

6. Evaluation Scheme:

Component	Duration	Marks (weightage %)	Date & Time	Remarks
Midsem Test	90 Mts.	75 (25 %)		CB/OB
Weekly Tutorial /Quiz	Continuous	60 (20 %)	Spread Across The Semester	OB
Lab Assignments /Design project/ term paper/ Quiz	(Continuous)	60 (20 %)	Spread Across The Semester	OB
Comprehensive Exam	3 Hours	105 (35%)		OB/ CB

7. **Chamber Consultation Hour:** To be informed by the respective tutorial instructor. To contact IC , fix a time. Email--anug@pilani.bits-pilani.ac.in

8. **Notices** regarding the course will be put up on **NALANDA** (LMS-e portal).

9. **Make-up Policy:** **Make-up** for mid-semester and comprehensive examination will be considered ONLY in genuine cases, and the IC decision will be final. Prior permission from the instructor-in-charge is necessary before the examination. (**Note: No make-up will be given for regular quizzes/, SPICE quiz/ test/ lab**).



- 10. Academic Honesty and Integrity Policy:** Academic honesty and integrity are to be maintained by all students throughout the semester, and no type of academic dishonesty is acceptable.
- 11. Minimum Marks Requirement:** A student should obtain 20% of the average of the class, or 10% of the course total marks, whichever is lower, to clear the course. If any student receives marks below the prescribed standard mentioned above, he/she may be awarded NC.

Instructor-in-Charge
EEE/INSTR F313